Engineering Chemistry (AHT-002) - Unit 3

Unit 3 - Water Chemistry & Corrosion - notes in easy English, matched to the UTU syllabus.

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What this unit covers (syllabus): hardness of water and its treatment (Clark's, lime-soda, zeolite, reverse osmosis), industrial water treatment, alkalinity, and corrosion - theories, types, mechanism and control.

Key ideas: Hardness: temporary (bicarbonates of Ca/Mg - removed by boiling or Clark's method with lime) and permanent (chlorides/sulphates - need lime-soda, zeolite or ion exchange). The **lime-soda process** adds $Ca(OH)_2$ and Na_2CO_3 to precipitate $CaCO_3$ and $Mg(OH)_2$; the **zeolite process** swaps Ca^{2+}/Mg^{2+} for Na^+ (regenerated with brine); **reverse osmosis** pushes water through a membrane against osmotic pressure. Hardness is expressed as $CaCO_3$ equivalents: mg/L = (mass of salt x 100 x 1000)/(MW of salt x volume). **Corrosion** = metal eaten by its environment: the **electrochemical theory** sets up tiny anode-cathode cells on the metal surface (rusting: Fe oxidises at the anode, O_2 reduces at the cathode). Types: galvanic, pitting, stress, water-line. Control: coatings, galvanisation (sacrificial zinc), cathodic protection, inhibitors.

Exam pointers: hardness numericals (convert to $CaCO_3$ equivalents, or lime/soda requirement) appear almost every year - practise the formula. "Explain the electrochemical theory of rusting with the anode/cathode reactions" and "compare lime-soda and zeolite" are the standard theory answers.

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